

Representative Structural Calculation Package

Torsion irregular corner frame

Item	Value
Project	Torsion irregular corner frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	10-torsion-irregular-frame

Nodes	Members	Loads	Combos	Max disp.	Max util.
88	179	504	28	6.754 mm	0.495

Model And Design Basis

Item	Value
Building type	irregular
Description	Five-story frame with a missing corner and offset stiffness distribution.
Occupancy	Office
Bounds X/Y/Z	15.00 / 15.00 / 17.00
Analysis status	OK
Design status	OK
Package audit	OK

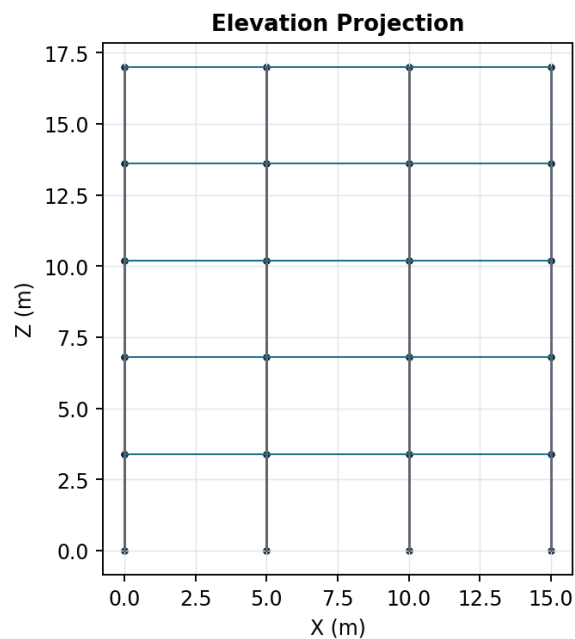
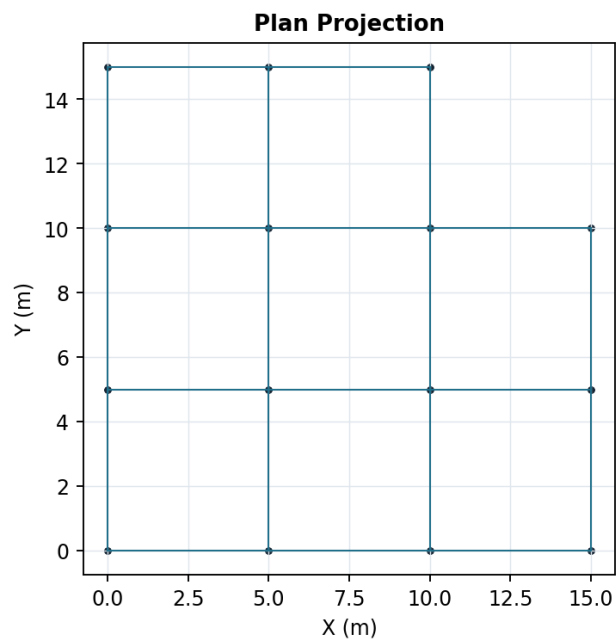
Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	106	0.000 kN	0.000 kN	-3712 kN
L	live	106	0.000 kN	0.000 kN	-1620 kN
WX	wind	73	114.75 kN	0.000 kN	0.000 kN
WY	wind	73	0.000 kN	114.75 kN	0.000 kN
EX	seismic	73	349.99 kN	0.000 kN	0.000 kN
EY	seismic	73	0.000 kN	349.99 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
2	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
3	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
4	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
5	225.00	742.50 kN	180.00 kN	22.95 kN	22.95 kN

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Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.155 mm	0.127	0.000
KDS-ST-02	OK	1.497 mm	0.172	0.000
KDS-ST-04-WX-P	OK	2.097 mm	0.187	0.000
KDS-ST-04-WX-N	OK	2.126 mm	0.186	0.000
KDS-ST-04-WY-P	OK	2.034 mm	0.186	0.000
KDS-ST-04-WY-N	OK	2.211 mm	0.186	0.000
KDS-ST-05-EX-P	OK	6.461 mm	0.268	0.000
KDS-ST-05-EX-N	OK	6.489 mm	0.269	0.000
KDS-ST-05-EY-P	OK	6.614 mm	0.270	0.000
KDS-ST-05-EY-N	OK	6.754 mm	0.267	0.000
KDS-ST-06-WX-P	OK	1.809 mm	0.120	0.000
KDS-ST-06-WX-N	OK	1.823 mm	0.120	0.000
KDS-ST-06-WY-P	OK	1.793 mm	0.120	0.000
KDS-ST-06-WY-N	OK	1.892 mm	0.119	0.000
KDS-ST-07-EX-P	OK	6.412 mm	0.201	0.000
KDS-ST-07-EX-N	OK	6.424 mm	0.202	0.000
KDS-ST-07-EY-P	OK	6.597 mm	0.203	0.000

KDS-ST-07-EY-N	OK	6.679 mm	0.201	0.000
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Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M67	OK	0.495	steel-deflection	KDS-ST-05-EY-N
2	M63	OK	0.494	steel-deflection	KDS-ST-05-EY-N
3	M71	OK	0.494	steel-deflection	KDS-ST-05-EX-P
4	M70	OK	0.486	steel-deflection	KDS-ST-05-EX-P
5	M66	OK	0.485	steel-deflection	KDS-ST-05-EY-N
6	M62	OK	0.483	steel-deflection	KDS-ST-05-EX-P
7	M73	OK	0.482	steel-deflection	KDS-ST-05-EX-P
8	M61	OK	0.475	steel-deflection	KDS-ST-05-EX-N
9	M60	OK	0.473	steel-deflection	KDS-ST-05-EY-N
10	M65	OK	0.473	steel-deflection	KDS-ST-05-EY-N
11	M69	OK	0.471	steel-deflection	KDS-ST-05-EX-N
12	M64	OK	0.470	steel-deflection	KDS-ST-05-EY-N
13	M72	OK	0.467	steel-deflection	KDS-ST-05-EX-N
14	M68	OK	0.464	steel-deflection	KDS-ST-05-EY-P
15	M53	OK	0.440	steel-deflection	KDS-ST-05-EY-N
16	M49	OK	0.439	steel-deflection	KDS-ST-05-EY-N
17	M57	OK	0.437	steel-deflection	KDS-ST-05-EX-P
18	M52	OK	0.430	steel-deflection	KDS-ST-05-EY-N
19	M56	OK	0.430	steel-deflection	KDS-ST-05-EX-P
20	M48	OK	0.429	steel-deflection	KDS-ST-05-EX-P

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	179	0	0.495
Connections	179	51	2.665
Foundations	15	51	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.

- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.